

Analysis of Vibrational Modes in the Context of Classical and Statistical Mechanics

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This paper explores the Fermi-Pasta-Ulam-Tsingou (FPUT) paradox, which challenges the assumption that nonlinearity in a system of coupled oscillators is sufficient to guarantee energy equipartition. We investigate the transition from non-ergodic to ergodic behavior by simulating a one-dimensional chain of masses connected by nonlinear springs. The system is studied using dynamical equations and the Verlet method. We demonstrate the existence of super-recurrence in the system and explore the critical initial energy per mass required for ergodic behavior as a function of the nonlinear spring constant and the number of masses. Our findings provide insights into the complex dynamics of the FPUT system and the conditions necessary for the onset of

I. INTRODUCTION

The foundational principles of classical mechanics and statistical mechanics have provided the frameworks for understanding the physical world, each from distinctly different perspectives. Classical mechanics allows for deterministic predictions of the motion of macroscopic objects under the influence of forces. On the other hand, statistical mechanics introduces a probabilistic approach, dealing primarily with the collective behavior of systems composed of a vast number of particles.

This research is motivated by the need to bridge the gap between these two fundamental theories, exploring their interface through a specific physical systems that exhibit behaviors not entirely explained by either framework alone. The system we are observing, specifically, is called the Fermi-Ulam-Pasta-Tsingou (FPUT) system, which features a lattice of masses connected by springs. This is supposed to simulate the interaction between vibrating molecules in a solid or liquid. Their ergodic behavior is predicted by statistical dynamics; however, in theory, the behavior should be completely deterministic. This paper will discuss how we can bridge this gap by introducing non-linearity in the springs; as well as get a more accurate simulation by increasing the lattice size.

II. METHODS

The system under investigation in this study is a one-dimensional chain of equally-spaced particles connected by springs. Instead of investigating the position of the masses directly, we describe these positions as a linear combination of the eigenmodes. We use the Euler-Cromer method to populate the first two time steps, then the Verlet method to numerically populate the next time steps.

A. System Description

The chain consists of N particles of equal mass m , where each particle is connected to its neighbors by springs with a spring constant k with the ends of the

chain fixed. In addition to the harmonic potential provided by the springs, a nonlinear force term is included to model interactions that go beyond the simple harmonic approximation. This non-linearity is crucial for exploring energy transfer mechanisms and thermalization processes that are central to the study.

Figure 1 is a schematic diagram of the one-dimensional chain model used in our simulations. The diagram illustrates the setup of particles connected by springs and the boundary conditions applied at the ends of the chain.

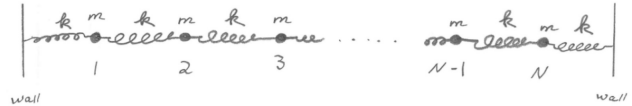


FIG. 1: Schematic diagram of the one-dimensional particle chain with fixed ends.

B. Dynamical Equation

The motion of each particle in the chain is described by the following differential equation:

$$m\ddot{x}_j = K(x_{j+1} - x_j) + \beta(x_{j+1} - x_j)^3 - K(x_j - x_{j-1}) - \beta(x_j - x_{j-1})^3, \quad (1)$$

where $\ddot{x}_i$ represents the acceleration of the j -th particle, and β is a coefficient that quantifies the strength of the nonlinear interactions.

C. Normalization of the Dynamical Equation

To simplify the analysis and focus on the fundamental behaviors of the system without the interference of specific units, we employ a normalization process on the dynamical equation. Normalization simplifies the equations and nondimensionalizes them, which is crucial for numerical simulations and theoretical analyses independent of specific unit scales.

The nondimensionalization process involves defining new dimensionless variables for position, time, and the

nonlinear spring constant β . These dimensionless variables, denoted as $\bar{x}$, $\bar{t}$, and $\bar{\beta}$, are defined in terms of characteristic quantities where $\bar{x} = \frac{x}{x_0}$ with $x_0 = 1$ representing the characteristic length scale, $\bar{t} = \frac{t}{t_0}$ with $t_0 = \sqrt{\frac{m}{k}}$ representing the characteristic time, and $\bar{\beta} = \frac{\beta}{\beta_0}$, where $\beta_0 = k$ is the characteristic scaling for the nonlinear spring constant.

Using these nondimensionalized variables, the original dynamical equation can be rewritten in normalized form by substituting $x = \bar{x}x_0$, $t = \bar{t}t_0$, and $\beta = \bar{\beta}\beta_0$. Given the definitions of x_0 , t_0 , and β_0 , the equation simplifies to:

$$\ddot{\bar{x}}_j = (\bar{x}_{j+1} - \bar{x}_j) + \bar{\beta}(\bar{x}_{j+1} - \bar{x}_j)^3 - (\bar{x}_j - \bar{x}_{j-1}) - \bar{\beta}(\bar{x}_j - \bar{x}_{j-1})^3. \quad (2)$$

This expression effectively sets $K = 1$ and $m = 1$ in the dynamical equation which lets us focus on just the effect of varying β .

For the remainder of this paper, we will omit the bars on the nondimensional variables for clarity and ease of reading. All quantities should be inferred to be normalized unless otherwise specified.

D. Numerical Integration Methods

As mentioned above, the Euler-Cromer method is used for the first two time steps, and the Verlet method is used for the next time steps.

The Euler-Cromer method, also known as the semi-implicit Euler method, is a variation of the Euler method that is particularly useful for solving second-order differential equations common in mechanics. It updates velocity before position, using the newly calculated velocity to update the position, which can lead to more stable and accurate simulations, especially for oscillatory motion. The method is outlined by the following equations:

$$v_{n+1} = v_n + a_n \Delta t \quad (3)$$

$$x_{n+1} = x_n + v_{n+1} \Delta t, \quad (4)$$

where v_n and x_n are the velocity and position at the n th timestep, a_n is the acceleration calculated from the forces at the n th timestep, and Δt is the timestep.

The Verlet method utilizes information of the position from a step steps prior as well as the current position and acceleration. The standard Verlet formulation is given by:

$$x_{n+1} = 2x_n - x_{n-1} + a_n \Delta t^2, \quad (5)$$

where x_n is the position at the n th timestep, x_{n-1} is the position at the previous timestep, and a_n is the acceleration at the n th timestep.

Both methods are second-order accurate, meaning they offer a good compromise between computational efficiency and accuracy in the integration of the equations of motion.

III. SYSTEM EQUATIONS

The following section regards equations used to calculate important quantities, such as, the location of each mass and the amplitude and energy of each mode.

The position $x_j(t)$ of the j -th mass at time t can be represented as a linear combination of all the normal modes of the system:

$$x_j(t) = \sqrt{\frac{2}{N+1}} \sum_{\ell=1}^N D_\ell(t) \sin\left(\frac{\pi \ell j}{N+1}\right), \quad (6)$$

where $D_\ell(t)$ represents the amplitude of mode ℓ at time t .

Given the position $x_j(t)$ determined by the Verlet method, the amplitude of each mode ℓ , $D_\ell(t)$, can be calculated using the formula:

$$D_\ell(t) = \sqrt{\frac{2}{N+1}} \sum_{j=1}^N x_j(t) \sin\left(\frac{\pi \ell j}{N+1}\right). \quad (7)$$

This formula provides a mechanism to isolate the contribution of each mode from the overall state of the system.

The energy of each mode ℓ at a given time t is calculated using the kinetic and potential energy contributions of the mode:

$$E_\ell(t) = \frac{1}{2} \left(\left(\frac{d}{dt} D_\ell(t) \right)^2 + (\omega_\ell D_\ell(t))^2 \right), \quad (8)$$

where ω_ℓ is the frequency of mode ℓ . This equation accounts for the dynamic nature of the energy distribution among the modes, critical for understanding the system's behavior over time.

IV. RESULTS

First, we can see the behavior of the system as predicted by classical mechanics. This is the completely ideal case, where the springs are assumed to be linear. Figure 2 shows a plot of the initial and final time, as well as the mode amplitudes and energies. From the plot, we can see that all the energy stays in mode 1. Equations 7 and 8 were used to calculate the amplitude and energy of each mode.

As we introduce a β term, the system displays a phenomena called super recurrence. Essentially, we must first define how to classify if a system is ergodic. We can say that if there is less standard deviation of energy among the modes compared to the initial, that it signifies the spread of energy from one mode to all the other modes. In the system shown by Figure 3, we introduce a small β . The bottom plot shows the standard deviation of mode energies initially start to rapidly decrease, but then given sufficient time, all of the energies return back to mode 1. This can be seen more cleanly in the top 2 plots of mode amplitudes and mode energies.

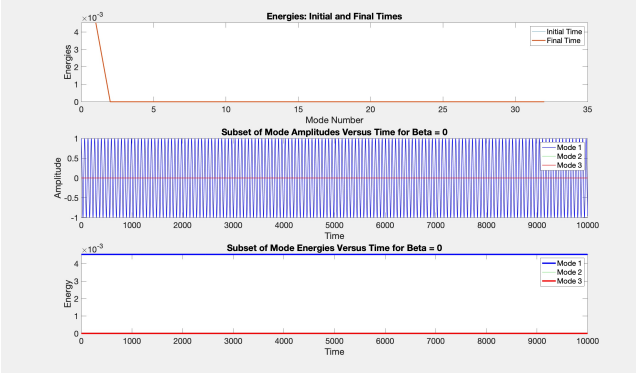


FIG. 2: System behavior when $\beta = 0$. The top plot shows energy in the modes at the initial and final time. The middle plot shows mode amplitudes versus time. The bottom plot shows mode energies versus time.

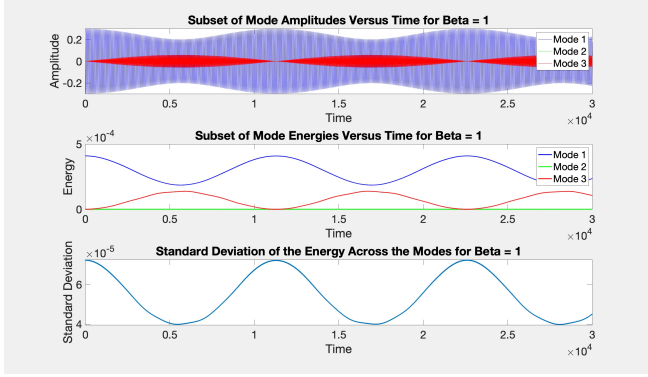


FIG. 3: System with $\ell_0 = 1$ showing super recurrence. The top plot shows mode amplitudes versus time. The middle plot shows mode energies versus time. The bottom plot shows standard deviation among mode energies versus time.

Next, we define a quantity $\epsilon = E_0/N$ that represents the initial energy per mass. We find that higher initial energy per mass makes the system easier to fall into ergodicity. This can be seen in Figure 4, where we find a ϵ_{crit} , which is the minimum ϵ that causes a system to go ergodic. The figure shows that a system with high initial energy per mass requires a lower β value to become ergodic.

Additionally, we expect that as $N \rightarrow \infty$, the system will behavior more realistically. We find that in Figure 5 that ϵ_{crit} rapidly decreases as N gets larger. For the systems in this plot, we also set the initial mode number to be $N/2$ so that the system's frequency can be roughly the same between each system.

We also see that super recurrence is not unique to just mode number 1. In Figure 6, we a similar pattern when the system is in mode number 5. One thing to note is that the energy first goes into the adjacent modes before returning to the original mode.

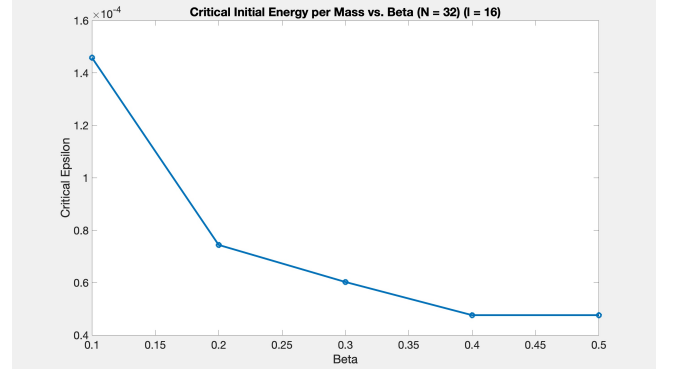


FIG. 4: Minimum initial energy per mass for given β to cause ergodicity.

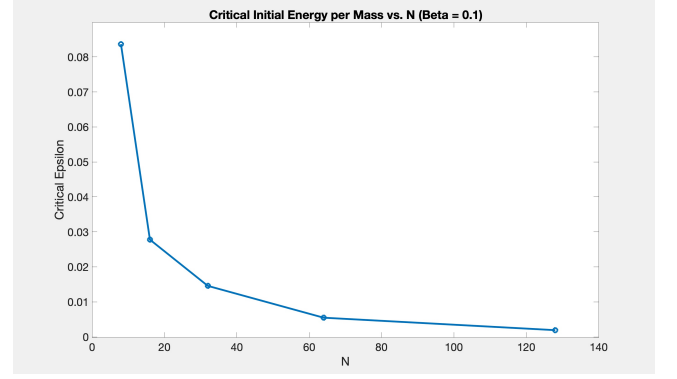


FIG. 5: Minimum initial energy per mass for given N to cause ergodicity.

V. DISCUSSION

Our initial experiments verify that energy injected into a single mode ($\ell = 1$) returns to that mode after a significant redistribution across other modes. This phenomenon, known as super-recurrence, was observed for multiple modes ($\ell \neq 1$), suggesting that this paradox is not unique to the lowest mode. However, the recurrence period and mode distribution are dependent on the non-linearity coefficient (β) and the initial energy per mass (ϵ). Also, we find that the energy will first enter into adjacent modes. This may be explained by continuity, as the frequencies present in the system change continuously over time. Because modes are listed in order of their frequency, then the transition between modes is also a continuous transition between frequencies, explaining why adjacent modes are affected first.

We explored the transition to ergodic behavior by varying ϵ . For each system configuration, we determined ϵ_{crit} , the minimum initial energy per mass required for the system to exhibit ergodic behavior. Our results indicate that as N increases, ϵ_{crit} decreases, implying that larger systems are more likely to reach ergodic behavior. This is consistent with the hypothesis that realistic physical systems (large N) behave according to statistical mechanics

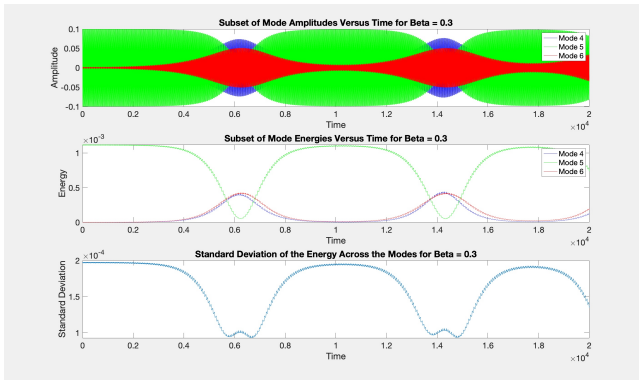


FIG. 6: System with $\ell_0 = 5$ showing super recurrence. The top plot shows mode amplitudes versus time. The middle plot shows mode energies versus time. The bottom plot shows standard deviation among mode energies versus time.

predictions.

While our study focuses on a one-dimensional chain, extending this work to two and three dimensions could provide further insight into the ergodic transition in nonlinear systems. Moreover, studying different forms of nonlinearity (e.g., 5-th order terms) can also provide a more realistic simulation of molecular dynamics.

VI. CONCLUSION

This study provides an investigation of the FPUT paradox and the transition from non-ergodic to ergodic behavior in a one-dimensional chain of masses connected by nonlinear springs. The observation of super-recurrence and the identification of a critical initial energy per mass for the onset of ergodicity contribute to our understanding of the complex dynamics of the FPUT system and the interface between classical and statistical dynamics.

The dependence of the critical initial energy per mass on the nonlinear spring constant and the number of masses highlights the interplay between system parameters and the emergence of ergodic behavior, specifically how departure from ideal and more realistic parameters lend itself to ergodicity. The decrease in the critical energy with increasing system size aligns with the expectation that larger systems should exhibit more realistic behavior.

In conclusion, this study provides a significant contribution to the understanding of the FPUT paradox and the conditions necessary for the onset of ergodicity in nonlinear coupled oscillator systems. The findings highlight the complex behavior of the FPUT system and serve as a foundation for further investigations into the fundamental principles governing the transition between classical mechanics and statistical mechanics.

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- [1] N. Giordano and H. Nakanishi, *Computational Physics* (Pearson Prentice Hall, Upper Saddle River NJ, 2006).